

# SilentKey Project Roadmap

## Vision

Build a portable identity platform where identity—not devices—is the primary trust anchor. Validate the protocol first; every feature depends on it.

## Current Status

Django initialized, PostgreSQL selected, QR generation using Segno completed, identity app created, admin under development.

## Development Philosophy

Protocol before implementation.

Business logic in Django.

Thin interfaces (CLI, Qt Desktop, Web, Mobile).

Performance modules in C++.

Persistent storage in PostgreSQL.

## Dependency Order

Identity → Device Registration → Pairing Session → Challenge → Authentication Session → Vault → Cross-device Sync → Community Sharing → P2P VPN.

## Phase 1

Finalize IdentityUser, UserDevice, PairingSession, AuthenticationChallenge and AuthenticationSession models. Implement Django Admin. Generate identity keypairs. Implement QR pairing (session identifier only). Implement challenge-response authentication. Audit every authentication.

## Phase 2

Implement encrypted vault, master key lifecycle, unlock flow, recovery planning and key rotation.

## Phase 3

Develop Qt desktop client that communicates exclusively through Django APIs. Keep business rules out of the UI.

## Research Areas

Biometric Cryptosystems, Fuzzy Extractors, Secure Sketch, Fuzzy Vault, Cancelable Biometrics, Threshold Cryptography, WebAuthn/FIDO2, Remote Attestation and Post-Quantum Cryptography.

## Important Architectural Decision

Treat biometric-derived cryptography as a research track. Complete the portable identity protocol first. Introduce a biometric abstraction layer later so future research does not require redesigning the platform.

## Immediate Sprint

1. Finish models. 2. Complete admin. 3. Pairing session. 4. Challenge generation. 5. Signature verification. 6. Audit logging. 7. CLI testing. 8. Repeat until protocol is stable.